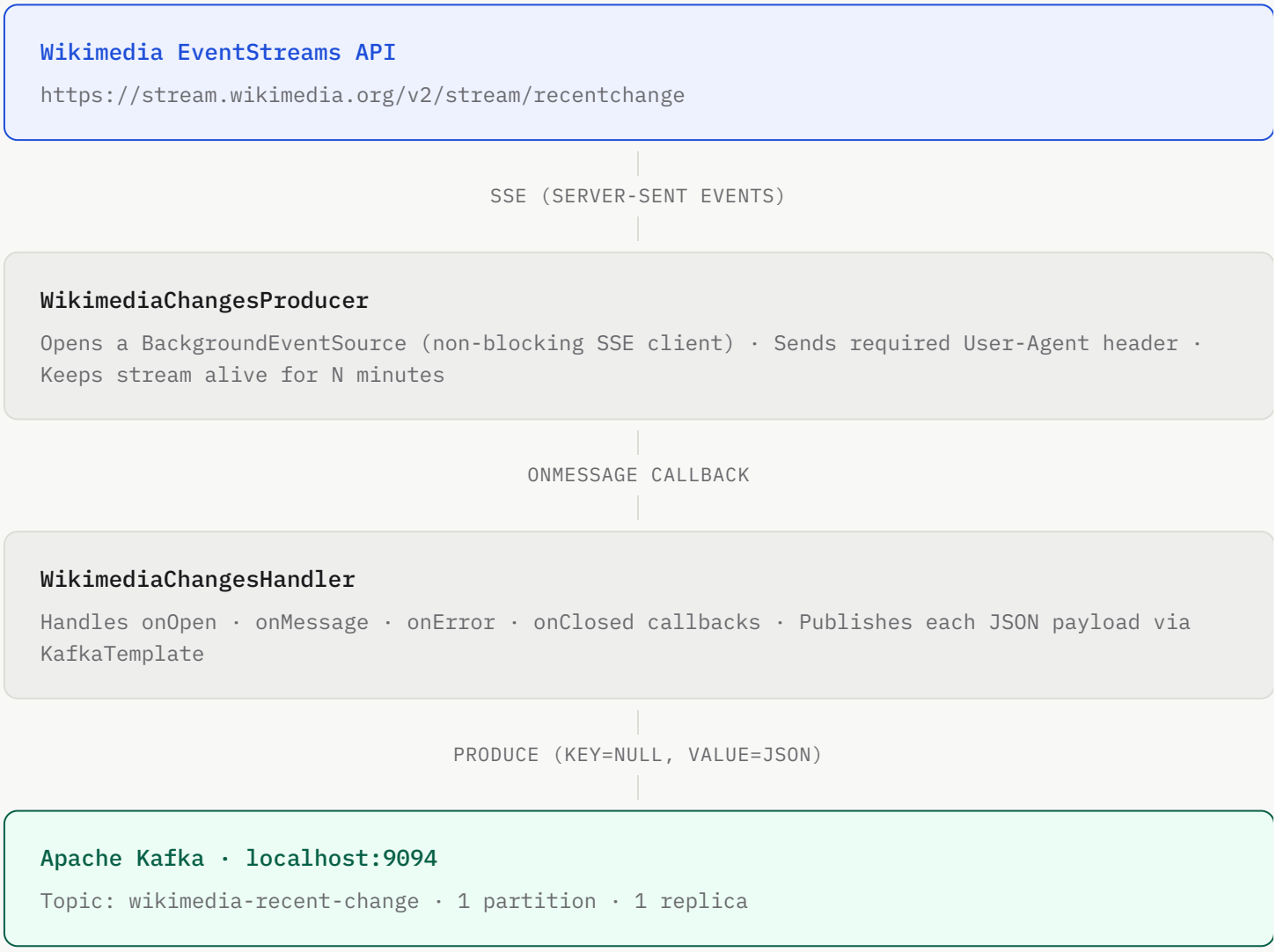
SPRING BOOT · APACHE KAFKA

Wikimedia Data Producer

Streams real-time Wikipedia edit events from the Wikimedia EventStreams API and publishes each change as a JSON message to an Apache Kafka topic.

Java 17 | Spring Boot 4.1.1 | Kafka Client 4.2.1 | okhttp-eventsource 4.3.0 | Lombok 1.18

01 Architecture



02 Package Layout

```
com.ms.wikimedia_data_producer
├── WikimediaDataProducerApplication.java // entry point; implements CommandLineRunner
│
├── config/
│   └── KafkaTopicConfig.java // declares the Kafka topic bean
│
└── stream/
    ├── WikimediaChangesProducer.java // opens & manages the SSE stream
    └── WikimediaChangesHandler.java // SSE callbacks → Kafka
```

CLASS	ROLE
`WikimediaDataProducerApplication`	Spring Boot entry point. Implements `CommandLineRunner` to kick off the stream immediately after the context starts.
`KafkaTopicConfig`	Declares a `NewTopic` bean so `KafkaAdmin` auto-creates the topic on startup if missing.
`WikimediaChangesProducer`	Builds a `BackgroundEventSource`, connects to the SSE endpoint with the correct User-Agent, and blocks the runner thread for the configured duration.
`WikimediaChangesHandler`	Receives every SSE callback. On `onMessage` it publishes the raw JSON payload to Kafka via `KafkaTemplate`.

03 Prerequisites

TOOL	MINIMUM VERSION	NOTES
Java (JDK)	17	Tested with Eclipse Temurin 17.0.20
Maven	3.8	Wrapper (`./mvnw`) is included
Apache Kafka	3.x / 4.x	Must be reachable at `localhost:9094`

START KAFKA WITH DOCKER

```
docker run -d \
  --name kafka \
  -p 9094:9094 \
  apache/kafka:latest
```

Or start the broker from the parent project's `docker-compose.yml`:

```
docker compose up -d
```

04 Configuration

All settings live in `src/main/resources/application.yaml`.

> ⚠ Wikimedia blocks requests that carry a generic or placeholder User-Agent. Replace the `contact:` value with your real email address before running.

PROPERTY	DEFAULT	DESCRIPTION
`spring.kafka.bootstrap-servers`	`localhost:9094`	Kafka broker address
`wikimedia.stream.url`	`https://stream.wikimedia.org/v2/stream/recentchange`	Wikimedia SSE endpoint
`wikimedia.stream.user-agent`	`wikimedia-data-producer/0.0.1 (contact: …)`	HTTP User-Agent — must contain a real email or project URL
`wikimedia.kafka.topic`	`wikimedia-recent-change`	Kafka topic to publish events to
`server.port`	`8081`	Embedded Tomcat port

05 How to Run

1 Build the project

Compile and package, skipping tests for a fast build.

```
mvn clean package -DskipTests
```

2 **Run in development (live reload)**

Spring Boot DevTools is on the classpath — changes to classes restart the context automatically.

```
mvn spring-boot:run
```

3 **Run the packaged JAR**

For a clean run outside the IDE or Maven.

```
java -jar target/wikimedia-data-producer-0.0.1-SNAPSHOT.jar
```

4 **Run tests**

```
mvn test
```

06 Runtime Flow

1 **Spring Boot starts on port 8081**

Application context initializes all beans.

2 **KafkaAdmin creates the topic**

If `wikimedia-recent-change` does not exist, it is created (1 partition, 1 replica).

3 **CommandLineRunner calls startStream()**

Executed immediately after the context is fully started.

4 **SSE connection opens to Wikimedia**

BackgroundEventSource connects with the configured User-Agent. The library reconnects automatically on transient errors.

5 **Each edit event is published to Kafka**

Raw JSON payload, no key (round-robin partitioning), idempotent producer enabled.

6 **Stream closes and app shuts down gracefully**

After 1 minute (configurable via `STREAM_DURATION_MINUTES`). The Kafka producer flushes pending messages before exit.

07 Verify Events Are Arriving

Use the Kafka CLI consumer to confirm messages are landing in the topic:

```
kafka-console-consumer.sh \
  --bootstrap-server localhost:9094 \
  --topic wikimedia-recent-change \
  --from-beginning
```

Each message is the raw JSON sent by the Wikimedia API, for example:

```
{
  "type": "edit",
  "wiki": "enwiki",
  "title": "Python (programming language)",
  "user": "SomeEditor",
  "timestamp": 1729033200,
  "bot": false,
  ...
}
```

08 Troubleshooting

HTTP error 403 from Wikimedia	Placeholder User-Agent in config	Set a real email in `wikimedia.stream.user-agent`
Connection refused to Kafka	Broker not running or wrong port	Start Kafka and confirm it listens on `localhost:9094`

Topic not created on startup	Broker unreachable when KafkaAdmin runs	Ensure the broker is fully up before starting the app
Stream disconnects and reconnects repeatedly	Transient network issue or Wikimedia rate-limit	Expected — the library retries automatically with backoff